# Andrew Shentu

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## EDUCATION

**University of California, Los Angeles** — B.S. Electrical Engineering Expected Dec 2027

GPA: 3.80 · Transferred from College of San Mateo (GPA: 3.97)

**Coursework:** Analog Electronic Circuits*, Digital Signal Processing*, Applied Numerical Computing*, Electromagnetics, Circuit Theory I & II, Signals & Systems, Digital Logic Design, Probability, Computer-Aided Design (SolidWorks), Data Structures in C++ **in progress*

## TECHNICAL SKILLS

**Hardware:** KiCad (schematic, layout, PCBA fab outputs), LTspice, SolidWorks (parts, assemblies, drawings), soldering, FDM 3D printing, laser cutting

**Firmware & Software:** C/C++, Python, MATLAB, QMK/VIA, Git

**Lab & Manufacturing:** Oscilloscope, multimeter, IEC-711 acoustic coupler, JLCPCB PCBA, overseas CNC sourcing

## EXPERIENCE

**Independent Keyboard Hardware Designer** — ShentoBento Mar 2026 – Present

- Designed two STM32F072-based keyboard PCBs in KiCad: a drop-in solder PCB for the Velvet Pro keyboard, developed with its designer (Ampersandy) from their board outline, and the Uni60, a universal 60% board.
- Extended the Uni60 to support ANSI and ISO, split backspace and shifts, stepped caps, and five bottom-row layouts; re-routed the matrix around tray-mount holes drilled between keys.
- Designed for cross-case compatibility with 5 daisy-chained JST-SH daughterboard positions aligned to different cases' cable channels, plus a Molex Pico-EZmate that sits lower than the switch pins and fits any case.
- Ran two PCBA production runs through JLCPCB (15 units in Apr 2026, 50 in Sep 2026), covering fab/assembly files, component sourcing, packaging and shipping; delivered 62 boards at cost to 26 customers.
- Flashed and tested every shipped board (QMK, row/column continuity from MCU to switch matrix); zero field failures reported to date. Open-sourced all PCB designs and QMK/VIA firmware.
- Designed a 6063 aluminum O-ring-mount 60% case with a brass weight in SolidWorks and sourced CNC machining and anodizing overseas (2-unit v1); for the 5-unit rev 2, moved the weight from 2 to 6 M3 screws to stop a ping caused by its unsupported span vibrating against the case.

## PROJECTS

**Tendon-Driven Bionic Hand** — UCLA engineering course project, team of 3 Fall 2025

- Designed the 3D-printed forearm and hand in SolidWorks, packaging 5 servos, an Arduino Uno and a battery pack; each servo spools a tendon that curls one finger, and all five fingers actuated in the final build.
- Arranged the five servo mounts so the forearm could still be assembled, routed the tendons along non-interfering paths, and shaped the tendon channels to print without collapsing.
- Set the finger joint geometry so the thumb and index finger meet at full closure, giving the hand a working pinch grip.

**Solar-Tracking Weather Station** — IDEA Hacks 2026 hackathon, team of 5 Apr 2026

- Owned the mechanical design: a laser-cut plywood enclosure and a servo-driven turntable that rotates the solar panel toward the sun.
- Designed the mounting for the team's four-sensor electronics (humidity, wind speed, solar irradiance, dust); built in 36 hours.

**Multi-Driver In-Ear Monitor Crossover** — personal project May 2026 – Present

- Designed a 4-way passive crossover PCB in KiCad (open-sourced as a template) that splits the audio signal across four balanced-armature drivers per side.
- Measured frequency response on an IEC-711 coupler and iterated RC values to shape the response toward a target curve.

## LEADERSHIP & ACTIVITIES

**IEEE at UCLA** — Student Project Initiative: $150 grant for my rev 2 multi-driver IEM (Oct 2026 – Present)

**Toastmasters District 4** — Area Director overseeing 3 clubs; named Area Director of the Year (2023)

**SCUDEM IX 2024** — Outstanding Award, differential-equations modeling competition

**First Presbyterian Church of Burlingame** — Audio-Visual Manager (2022–2024)